Total-fit line breaking in narrow measures

Ada Example[1] and Ben Sample[2]

[1] Department of Typography, Example University
[2] Institute for Document Engineering, Sample College

Abstract. A line breaker that looks at the whole paragraph at once finds better breaks than one that fills each line as far as it can. The advantage grows as the measure gets narrower, because a narrow line has fewer spaces to absorb the leftover width. We restate the box, glue and penalty model, show how badness reacts to stretching and compare first-fit and total-fit breaking on a set of measures typical for two-column layouts. The numbers in this example document are illustrative.

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1 Introduction

Justified text distributes the leftover width of a line over its interword spaces. A greedy breaker decides line by line: it puts as many words on a line as fit and moves on. The total-fit algorithm of Knuth and Plass [1] instead treats the paragraph as a whole and picks the sequence of breakpoints with the least total cost. It has been the line breaker of TeX since 1982 [2].[1]

Narrow measures, such as the columns of a journal page, are where the choice matters most. Section 2 recalls the model, Section 3 compares the two strategies.

2 The box, glue and penalty model

A paragraph is a sequence of boxes, glue and penalties (Figure 1). A box has a fixed width, a glyph or a whole word. Glue has a natural width that can stretch by at most y and shrink by at most z. A penalty marks a possible break and its cost.

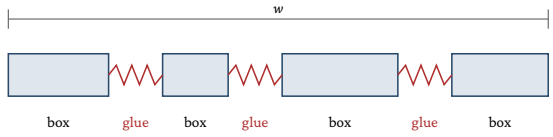


Figure 1. A line of width w: boxes keep their width, glue stretches or shrinks to fill the line.

For a line with natural width x, total stretchability y and measure w, the adjustment ratio is

$$r = \frac{w - x}{y} \tag{1}$$

where z takes the place of y when the line has to shrink. The badness of the line grows with the cube of the ratio,

$$b = 100|r|^3 \tag{2}$$

and the demerits of a break combine the badness with the penalty p of the breakpoint and a constant line penalty l:

$$d = (l + b)^2 + p^2 \quad (p \geq 0) \tag{3}$$

The algorithm minimizes the sum of the demerits over all lines of the paragraph, see Equation (3).

3 Narrow measures

The cube in Equation (2) makes badness rise steeply once a line stretches beyond its natural stretchability (Figure 2). A line with $r = 1.3$ already has a badness of about 220 and is rejected under the tolerance of 200 that plain TeX uses.

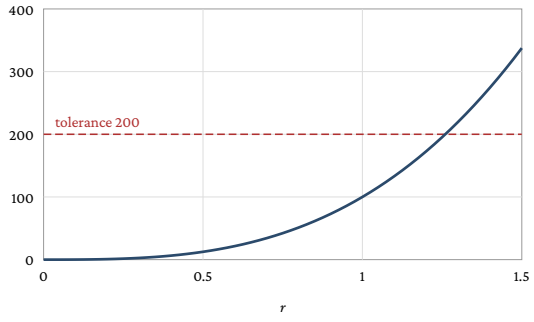


Figure 2. Badness as a function of the adjustment ratio. The dashed line marks the tolerance of 200.

Table 1 lists the outcome for four measures, given in ems of the text font. Total-fit keeps the average badness low down to the narrowest measure, where

1. The document you are reading is set by glu, which uses the same algorithm through the boxes and glue library.

first-fit leaves loose lines that a reader notices as rivers of white space.

Table 1. Average badness b and loose lines (b > 200) per 100 lines, first-fit (FF) against total-fit (TF). Illustrative values.

Measure	FF b	TF b	FF loose	TF loose
35 em	18	9	1	0
28 em	31	14	3	0
22 em	57	23	8	1
16 em	112	41	19	4

Hyphenation widens the gap further: it adds breakpoints, and only an algorithm that weighs them against each other can use them without hyphenating every other line. Plass extended the same idea from lines to pages [3]; the CSS multi-column module [4] describes how columns are filled, but leaves the line breaker to the implementation.

4 Conclusion

Breaking a paragraph as a whole costs more computation than breaking it line by line, but the cost is small for a modern machine and the gain is largest exactly where journals set their text: in narrow columns.

References

[1] D. E. Knuth and M. F. Plass. Breaking paragraphs into lines. *Software: Practice and Experience*, 11(11):1119–1184, 1981.

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[3] M. F. Plass. *Optimal Pagination Techniques for Automatic Typesetting Systems*. PhD thesis, Stanford University, 1981.

[4] W3C. *CSS Multi-column Layout Module Level 1*. W3C Candidate Recommendation.